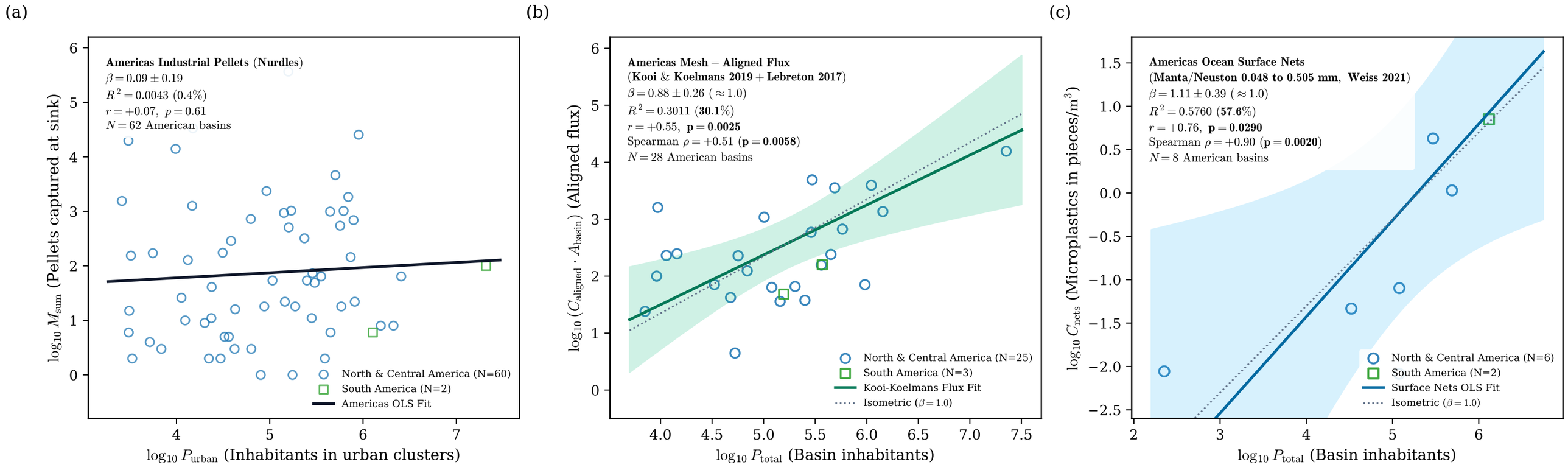

(a)
Americas Industrial Pellets (Nurdles)
$\beta = 0.09 \pm 0.19$
$R^2 = 0.0043$ (0.4%)
$r = +0.07$, $p = 0.61$
$N = 62$ American basins
North & Central America (N=60)
South America (N=2)
Americas OLS Fit
$\log_{10} M_{\text{sum}}$ (Pellets captured at sink)
$\log_{10} P_{\text{urban}}$ (Inhabitants in urban clusters)
(b)
Americas Mesh – Aligned Flux
(Kooi & Koelmans 2019 + Lebreton 2017)
$\beta = 0.88 \pm 0.26$ (≈ 1.0)
$R^2 = 0.3011$ (30.1%)
$r = +0.55$, $\mathbf{p = 0.0025}$
Spearman $\rho = +0.51$ ($\mathbf{p = 0.0058}$)
$N = 28$ American basins
North & Central America (N=25)
South America (N=3)
Kooi-Koelmans Flux Fit
Isometric ($\beta = 1.0$)
$\log_{10}(C_{\text{aligned}} \cdot A_{\text{basin}})$ (Aligned flux)
$\log_{10} P_{\text{total}}$ (Basin inhabitants)
(c)
Americas Ocean Surface Nets
(Manta/Neuston 0.048 to 0.505 mm, Weiss 2021)
$\beta = 1.11 \pm 0.39$ (≈ 1.0)
$R^2 = 0.5760$ (57.6%)
$r = +0.76$, $\mathbf{p = 0.0290}$
Spearman $\rho = +0.90$ ($\mathbf{p = 0.0020}$)
$N = 8$ American basins
North & Central America (N=6)
South America (N=2)
Surface Nets OLS Fit
Isometric ($\beta = 1.0$)
$\log_{10} C_{\text{nets}}$ (Microplastics in pieces/m^3)
$\log_{10} P_{\text{total}}$ (Basin inhabitants)